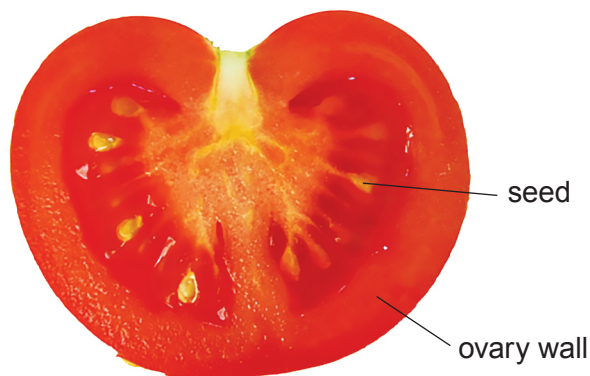


3. Flowers may be insect or wind-pollinated. After pollination, plants produce seeds. The photographs below show a tomato flower, which is insect-pollinated, and a vertical section through a tomato fruit.



- (a) State how the appearance of the flower would be different if the tomato flower was wind-pollinated. [2]

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- (b) The ovary wall of the tomato fruit produces chemicals that inhibit germination of seeds before they are dispersed.

An experiment was carried out to compare the concentration of inhibitor in two varieties of tomato. Germination inhibitors were extracted from the ovary wall as follows:

- grind up tissue in solvent using a pestle and mortar
- filter the extract
- place extract on filter paper
- allow solvent to evaporate

- (i) State **three** ways in which you would ensure that the extracts from both varieties of tomato were comparable. [2]

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To compare the concentration of inhibitor the following procedure was followed:

- wash seeds in distilled water
- place seeds on filter paper containing extract
- provide the conditions required for germination
- calculate the percentage of seeds germinated after three days

(ii) Explain why the seeds need to be washed before placing on the filter paper. [1]

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(iii) Explain why the extracts from both varieties of tomato were tested on seeds from both varieties. [1]

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(iv) Describe and explain a suitable control for this experiment. [2]

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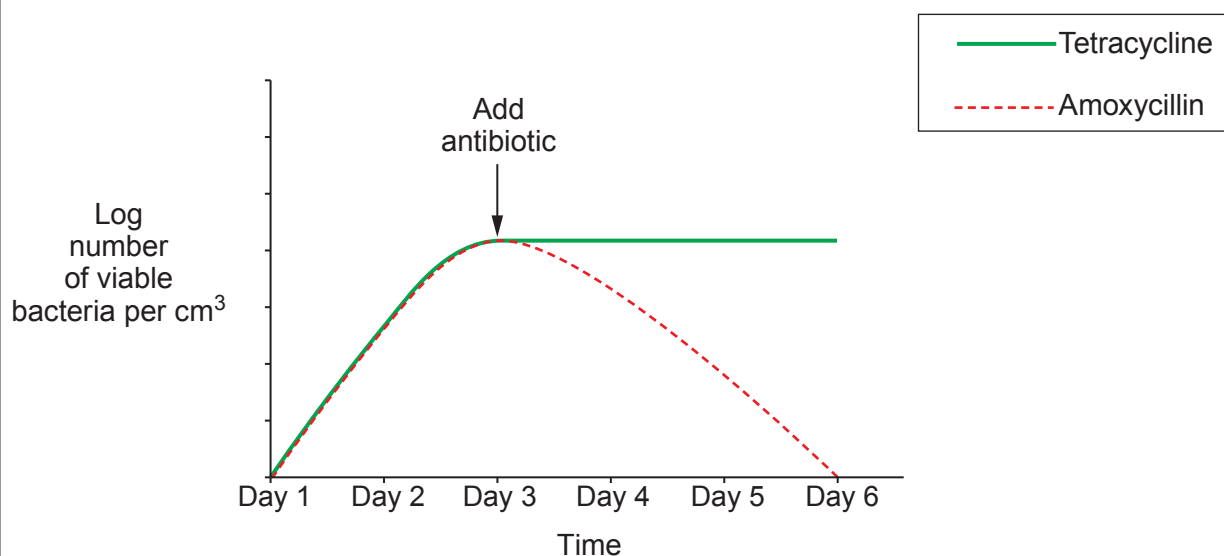
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8



Scientists studied the effect of tetracycline and amoxycillin on *S. pyogenes*.

The same number of bacteria were inoculated into the same type of nutrient medium and cultured for 6 days at 35 °C in aerobic conditions. Amoxycillin was added to one culture and tetracycline was added to the other at day 3. The graph below shows how the number of viable bacteria change during the course of the experiment.



- (iii) Using the graph, identify the mode of action of these two antibiotics and explain how you reached this conclusion. [2]

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- (iv) State **two** factors that would need to be controlled to increase confidence in the results. [2]

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- (v) Explain why *S. pyogenes* was cultured at 35 °C. [1]

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(c) Vaccination programmes for the common cold caused by rhinovirus have so far proved unsuccessful.

- (i) Using your own knowledge of immunisation, suggest and explain why it is unlikely that a vaccine for the common cold will ever be developed. [3]

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- (ii) The flu virus causes many similar symptoms to the common cold. There is a vaccine against the flu virus which can provide protection. It is offered to certain 'at-risk' groups. These immunisations are not compulsory.

Suggest why it would be considered unethical to make the flu vaccine compulsory for all. [1]

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